Calculate gravitational force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate a gravitational interaction with consistent units.

SUCCESS CRITERIA - I can...

1. Identify both masses and the center separation.
2. Square the distance before dividing.
3. Check units and the scale of the result.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “separation” and “mass” to explain your model or result. Sentence starters are optional.

separation: distance between the specified objects or centers

mass: a measure of how strongly an object resists acceleration

inverse square: proportional to one divided by distance squared

01 Warm-up: bring back an earlier idea

A ball begins with downward velocity 3 m/s and falls 2 s.
Use downward positive, g=10 m/s² and no drag.
Find final velocity and displacement.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

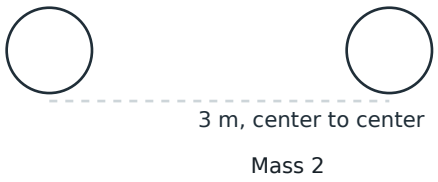
02 Worked example: explain the model

Reference: $F = Gm_1m_2/r^2$; $G = 6.67\times10^{-11}$ N·m²/kg².
r is center-to-center separation. CPA may use the supplied numerator.

Worked reasoning: In a supplied gravity model, $Gm_1m_2=18$ N·m² and r=3 m. Square first: $r^2=9$ m², so $F=18/9=2$ N, attractive.

Your turn: Sketch the two centers, label r and add the two force arrows.
Show how the units reduce to newtons.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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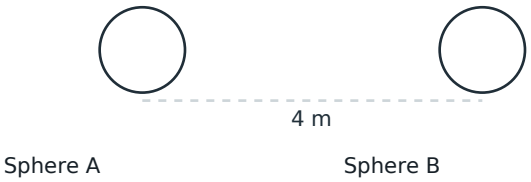
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

For spheres 4 m apart center to center, a student substitutes r=2 m because each is 2 m from the midpoint. **Identify** the error.

Sentence starter: The step I would change is … . The evidence is … .



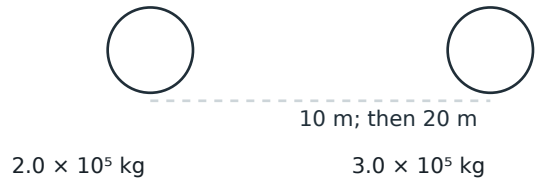
Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: 1 2 3

04 Apply the physics

Two idealized spherical space objects have masses 2.0×10^5 kg and 3.0×10^5 kg. Their centers are 10 m apart. **Use** $G = 6.67 \times 10^{-11}$ N·m²/kg². **Label** the model and calculate their attraction. Repeat at 20 m.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

Why can a tiny force between classroom-sized objects be physically real even if you cannot feel it?

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A 4 kg box slides left at 6 m/s on a rough horizontal floor after release.
Draw forces and explain what follows about vertical and horizontal acceleration.
Draw the forces on the chosen object and label their sources.

Sentence starter: Under the assumption …, the evidence … supports … because … .



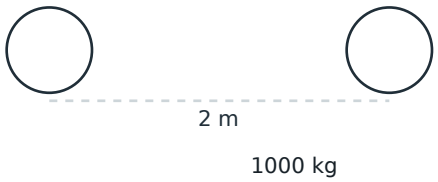
Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

Two 1000 kg spheres have centers 2 m apart.
Use $G = 6.67 \times 10^{-11}$ N·m²/kg².
Calculate the attraction. A classmate divides by r instead of r²: by what factor does that change the numerical answer?

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

08 STRETCH: test the model

A pair has force 0.12 N at 2 m. Without finding either mass, calculate Gm_1m_2 and the separation for a force of 0.03 N.

Sentence starter: I found … [unit]. This value means … in this situation.

Separation (m)	Force (N)
___	___
___	___

Marzano check - circle: ① ② ③